

Application of Schrödinger's equation in a box using a single, double and triple slit configuration

Julián Guerrero Cánovas, María Pérez Martínez, Elena Pujante Martínez and Iván Villegas Pérez
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In this report we have simulated a double slit configuration in a box using Schrödinger's equation, specifically, we have looked at the time evolution of the probability of detecting a particle at a certain position (x, y) . In order to do that, we have used a Gaussian wave function for the initial state of the particle. For the already mentioned configuration, we have studied as well the detection probability when spanning the entire y axis at a certain x position at a given time. This has also been done for the single and triple slit configurations. We have been able to see the expected diffraction, explained by Huygens' principle, as well as the precision of the Crank-Nicolson method.

I. INTRODUCTION

The double-slit experiment demonstrates that light and matter can display characteristics of both classically defined waves and particles. It also displays the fundamental probabilistic nature of quantum mechanical phenomena [1].

We will simulate the two-dimensional time-dependent (2 space dimensions and 1 time dimension) Schrödinger's equation (Equation 1), and use it to study a double-slit in a box experiment and some variations of it, as the single and triple slit configurations. The main purpose of this report is to solve this differential equation using an approximation method. In order to study the effectiveness of the approach used, we will study the deviation of the total probability from one (the exact total probability) and the evolution of the detection probability with time. This last section will also allow us to picture how non-relativistic particles interact with a determined n-slit configuration.

In section II we describe the mathematical background for our system and we formulate a concrete algorithm, the Crank-Nicolson algorithm (described in Algorithm 1), which can be implemented in any programming language¹ to pursue our main objectives: study the effectiveness of this approach to illustrate how photons behave at a double-slit-in-a-box setup and variations thereof. In section III we present and discuss in detail the obtained results related to single, double and triple slit configurations. For a double-slit configuration, we show the time evolution of the 2D probability function which denotes

the probability density for detecting the particle at a certain position at different times; and the detection probability spanning the entire y axis at a certain position in the x axis at a certain time. We also show this last detection probability for single and triple slit configurations. Finally, in section IV we provide a short summary and outlook of all of our work.

II. METHODS

Background

As we have already said in section I, we will start by introducing the mathematical and physical background at which our system is subjected.

The general formulation of the time-dependent Schrödinger's equation is

$$i\hbar \frac{d}{dt}|\Psi\rangle = \hat{H}|\Psi\rangle, \quad (1)$$

where $\hat{H}$ is some Hamiltonian operator and $|\Psi\rangle$ is a quantum state. The quantum state is related to probability through the Born rule. The Born rule is a postulate which states that the probability density of finding a particle at a given point, when measured at time t , is proportional to the product of the wave function and its conjugate at that point.

In the case of a single, non-relativistic particle in two dimensions, if we work in "position space", the quantum state $|\Psi\rangle$ can then be expressed as a complex valued function $\Psi(x, y, t)$, i.e. the wave function. Schrödinger's equation then becomes

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, t) =$$

¹ We have implemented this algorithm in C++ and the plots have been generated with Python. Every program we have used can be found in our public [GitHub repository](https://github.com/ivanvillegas7/Project-5) with the following link: <https://github.com/ivanvillegas7/Project-5>

$$= -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \Psi(x, y, t) + V(x, y, t) \Psi(x, y, t),$$

where the terms $-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2}$ and $-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial y^2}$ are expressions of the kinetic energy, corresponding to $\frac{p^2}{2m}$ from classical physics, with m being the particle mass. The external environment is encoded in the potential V . We will only consider the case of a time-independent potential, i.e. $V = V(x, y)$.

In this position space the Born rule takes the form

$$p(x, y; t) = |\Psi(x, y, t)|^2 = \Psi^*(x, y, t) \Psi(x, y, t),$$

where $p(x, y; t)$ denotes the probability density for detecting the particle at position (x, y) if we perform a position measurement at time t .

If all dimensionful variables are scaled away, i.e. all the variables are dimensionless, Schrödinger's equation can be rewritten in the following form:

$$i \frac{\partial u}{\partial t} = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + V(x, y) \cdot u. \quad (2)$$

In our new notation the Born rule takes the form

$$p(x, y; t) = |u(x, y, t)|^2 = u^*(x, y, t) u(x, y, t), \quad (3)$$

assuming the wave function $u(x, y, t)$ has been properly normalized.

In order to set up our wave equation, we have assumed Dirichlet boundary conditions in the XY plane:

- $u(x = 0, y, t) = 0$
- $u(x = 1, y, t) = 0$
- $u(x, y = 0, t) = 0$
- $u(x, y = 1, t) = 0$

For the initial wave function $u(x, y, t = 0)$ we will use a quantum mechanical Gaussian wave packet. An unnormalised Gaussian wave packet can be represented with the following expression

$$u(x, y, t = 0) = e^{-\frac{(x-x_c)^2}{2\sigma_x^2} - \frac{(y-y_c)^2}{2\sigma_y^2} + ip_x(x-x_c) + ip_y(y-y_c)}.$$

where x_c and y_c are the coordinates of the centre of the initial wave packet; σ_x and σ_y are the initial widths of the wave packet in the x and y directions, and p_x and p_y are the wave packet momenta. The initial state satisfies the boundary conditions and has been normalised such that

$$\sum_{i,j} u_{ij}^{0*} u_{ij}^0 = 1,$$

i.e. that the total probability in our 2D probability function $p_{ij}^n = u_{ij}^{n*} u_{ij}^n$ starts out normalized to 1. p_{ij}^n is the probability that a time step n is associated with a small grid cell of area h^2 centered on $(x, y) = (x_i, y_j)$

Algorithm 1 Crank-Nicolson

$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{F_i^{n+1} - F_i^n}{2}$	▷ Do the discretization with the Crank-Nicolson approach.
$A\vec{u}^{n+1} = B\vec{u}^n$	▷ Get matrix equation
$\vec{b} = B\vec{u}^n$	▷ Solve the matrix equation
$A\vec{u}^{n+1} = \vec{b}$	

The algorithm

Now that we have stated the environment in which our system develops and what expressions represent it, let's take a look on the approach used to evolve the system in time.

The Crank-Nicolson method is a finite difference method used for numerically solving partial differential equations (PDEs from now on).

The approximation of derivatives by finite differences is quite important when obtaining the numerical solution of differential equations, especially boundary value problems. The forward difference scheme approximates $\frac{\partial u}{\partial t}$ by the finite difference $\frac{u_i^{n+1} - u_i^n}{\Delta t}$ while the backward difference scheme approximates $\frac{\partial u}{\partial t}$ by $\frac{u_i^n - u_i^{n-1}}{\Delta t}$.

If we name $\frac{\partial u}{\partial t} = F(x, t)$, then $\frac{u_i^{n+1} - u_i^n}{\Delta t} = F_i^n$ and $\frac{u_i^{n+1} - u_i^n}{\Delta t} = F_i^{n+1}$. The linear combination of these two schemes results in $\frac{u_i^{n+1} - u_i^n}{\Delta t} = \theta F_i^{n+1} + (1 - \theta) F_i^n$ where $\theta \in [0, 1]$. This θ rule provides a family of finite difference approximations in time. $\theta = 0$ gives the Forward difference scheme in time, $\theta = 1$ gives the Backward difference scheme in time and the combination of the backward and forward difference methods, i.e. $\theta = 1/2$, gives the Crank-Nicolson scheme in time [2].

The basic idea behind this algorithm is to approximate the numerical derivative of $u(x, y, t)$ with a lower global error than both the Forward and Backward Euler schemes, by discretizing the mentioned derivative.

Now we can apply the Crank-Nicolson approach to our particular case, discretizing Equation 2. After some steps which can be seen in the appendix (section V) we obtain:

$$u_{i,j}^{n+1} - r(u_{i+1,j}^{n+1} - 2u_{i,j}^{n+1} + u_{i-1,j}^{n+1})$$

$$-r(u_{i,j+1}^{n+1} - 2u_{i,j}^{n+1} + u_{i,j-1}^{n+1})$$

$$+ \frac{i\Delta t}{2} V_{i,j} u_{i,j}^{n+1} = u_{i,j}^n - r(u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n)$$

$$-r(u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n) - \frac{i\Delta t}{2}V_{i,j}u_{i,j}^n,$$

where $r \equiv \frac{i\Delta t}{2h^2}$.

When taking into account the boundary conditions, the last equation can be expressed as the following

$$A\vec{u}^{n+1} = B\vec{u}^n, \quad (4)$$

where the matrices A and B can be also found in section V and where vector $\vec{u}^n$ (and similarly $\vec{u}^{n+1}$) is a column vector which contains the u_{ij}^n values for all the internal points of the XY grid at time step n . In row-form it would look like this:

$$\vec{u}^n = \begin{bmatrix} u_{1,1}^n \\ u_{2,1}^n \\ \cdot \\ \cdot \\ u_{M-2,1}^n \\ u_{1,2}^n \\ u_{2,2}^n \\ \cdot \\ \cdot \\ u_{M-2,2}^n \\ \cdot \\ \cdot \\ \cdot \\ u_{1,M-2}^n \\ \cdot \\ \cdot \\ \cdot \\ u_{M-2,M-2}^n \end{bmatrix}$$

where the parentheses are just added to make it clear where there is a change in the second index j . Thus the $\vec{u}$ vectors have length $(M-2)^2$, and the matrices A and B are matrices of size $(M-2)^2 \times (M-2)^2$.

There are various approaches to solve Equation 4, for instance the Jacobi method, the Gauss-Seidel and the Successive over-relaxation (SOR) method. The approach which might be best well-suited to solve this matrix equation is the Gauss-Seidel method since it updates/overwrites the value of a variable as soon as a new value is available. The Jacobi method uses all the values obtained in the previous iteration in the next one, so all these values need to be retained to perform the next iteration. In our case, retaining all the values would make our code inefficient since we are working with high dimension matrices. The Gauss-Seidel iterative method is a variant of the Jacobi method which improves the rate of convergence. Both of these methods always converge if the matrix is diagonally dominant, which is our case.

In our program we have simply used the already implemented command “spsolve”, but any of these approaches would be equally good.

III. RESULTS AND DISCUSSION

The total probability in our 2D probability function $p_{ij}^n = u_{ij}^{n*} u_{ij}^n$ should be conserved over time. We have tested this by running two simulations of the double-slit-in-a box setup for the cases without and with a double-slit barrier. The main simulation parameters are h , Δt , T , x_c , σ_x , p_x , y_c , σ_y , p_y and V_0 . The values of these parameters that have been considered for both simulations are the following: $h = 0.005$, $\Delta t = 2.5 \times 10^{-5}$, $T = 0.008$, $x_c = 0.25$, $\sigma_x = 0.05$, $p_x = 200$, $y_c = 0.5$, and $p_y = 0$. The values of σ_y and V_0 are different, for the first case $\sigma_y = 0.05$ and $V_0 = 0$ whereas for the second case $\sigma_y = 0.10$ and $V_0 = 1 \cdot 10^{10}$. The double-slit configuration used consists of a wall of thickness in the x direction of 0.02, a wall position (centre) in the x direction of 0.5, a length of the wall piece separating the two slits (the y distance between the inner edges of the two slits) of 0.05 and a slit aperture (opening in the y direction) of 0.05.

In Figure 1, it can be seen as a function of time the deviation of the total probability for both simulations. For the case without a slit, it could be said that the total probability does not deviate from the exact one whereas in the case with a double slit it deviates a little from one. In theory the total probability should be one, and should be conserved over time, but, in our case, the deviation is due to lack of precision in the method used to approximate the result of the Equation 1. In the case of no slit, i.e. the potential is 0, there is a term less in the equation mentioned before and therefore less error, which matches with what is observed in 1.

We have used the double-slit configuration with the same settings as the second simulation mentioned before to plot some colourmaps which illustrate the time evolution of the 2D probability function $p_{ij}^n = u_{ij}^{n*} u_{ij}^n$ showing here the times steps $t = 0$, $t = 0.001$ and $t = 0.002$ respectively, as can be seen in Figure 2. In the first plot, the 2D probability for detecting the particle at position (x, y) shows that the particle has not gone through the slits yet, being this probability centered, as expected from the simulation parameters, at $x = x_c$ and $y = y_c$ with widths in the x and y directions of δ_x and δ_y respectively. In the second one, the particle is going through the slit, and a phenomenon of diffraction happens. For a better understanding of this phenomenon (more often used in optics) we can take a look at the Huygens principle. Huygens conceived light as a wave propagating through a medium (as it is, more or less, our case). He put forward the hypothesis that every point on a propagating wavefront becomes

the source of a secondary spherical wave emitted from each point of the medium. The combination of all these secondary waves is the shape of a wavefront at a later time. In fact, looking at the third plot in Figure 2, one can see how these “secondary waves” form a new wave front at a later time. The reflected wave that we also observe in this third plot, is the result from the sum of two waves. The first one comes from the reflection that occurs between the incoming wave and the wall. The second one is the propagating-to-the-left part of the wave resulting from the already mentioned diffraction.

For the same time steps we have also plotted colourmaps that show $\text{Re}(u_{ij})$ and $\text{Im}(u_{ij})$ (Figures 3 and 4). With these plots we can see in a more clear way how the wavefronts behave. As we have already said, in the plots for $t = 0$, plane wavefronts are traveling towards the double slit wall. In the plots corresponding to $t = 0.001$, it can be seen that the part of these wavefronts which collides with the wall reverse their propagation direction, but stay with the plane wave form. The segment which “collides” with the slits suffers diffraction and part of it continues propagating in the same direction while another part reverses its propagation direction, forming both spherical wavefronts. All of this phenomena can be clearly seen in the last plots of both figures (Figures 3 and 4). In the right part of the plots, the reflected plane waves resulting from the collision with the wall, interfere with the spherical waves resulting from the diffraction caused by the slits. In the left part, we can see that the refracted spherical waves, also caused by the diffraction at the slits, have a bigger amplitude than the reflected ones caused by the same phenomenon.

We have separated the real and the imaginary parts because it is difficult to plot and interpret them together, but we can see that they behave in more or less the same way.

It is useful to also take a look on the complete time evolution of the 2D probability function, $\text{Re}(u_{ij})$ and $\text{Im}(u_{ij})$ so here we attach an animation of each of these processes: [2D probability](#), [Re\(\$u_{ij}\$ \)](#) and [Im\(\$u_{ij}\$ \)](#).

For a single, double and triple slit configuration we have shown the detection probability spanning the entire y axis at a certain position in the x axis ($x = 0.8$) at a certain time ($t = 0.002$) as it is shown in Figure 5. The effect of having more slits implicates that the probability of finding the particle at some position y is essentially zero except in distinct points. It can be observed as well that, as more slits are introduced in the configuration, more peaks of bigger amplitude appears. The position of the peaks will depend on the slit separation and aperture and on the wavelength.

In this case, it is really useful to compare that figure

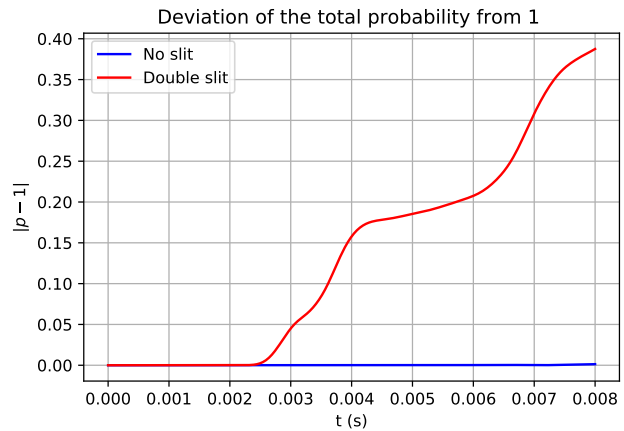


FIG. 1. Deviation of the total probability from 1.0 as a function of time for the cases with and without a double-slit barrier

(Figure 5) to the one representing the intensity at Thomas Young’s double-slit experiment. In fact, the probability is proportional to square modulus of the amplitude of the wave, i.e. $|u|^2$ and the intensity is proportional to the square modulus of the electric field (also represented by a wave). Thus, we can say that a correspondence between the classical intensity and the quantum probability exists. In fact, light can be understood as a collection of photons, mass-less particles that also follow the quantum laws that we have studied in this report.

IV. CONCLUSION

In this report, the passages of a non-relativistic particle through a single, double and triple slit configuration in a box have been simulated. We have done this in order to study the effectiveness of approximating the solution to a version (version in which all dimensionful variables have been scaled away and a time-independent potential has been taken into account) of the Schrödinger equation (Equation 2) using the Crank-Nicolson algorithm. In addition to that goal, we wanted to observe as well the time evolution of the probability of detecting a particle at a certain position (x, y) .

Firstly, we have simulated two cases, one without a slit and another one with a double-slit configuration (Figure 1). It has been observed the deviation of the total probability of detecting the particle for both simulations, being the reason for that deviation the lack of precision in the Crank-Nicolson method to approximate Equation 1. Nevertheless, the value of this deviation is quite small, therefore we can affirm that the approach used is really useful to get an approximate solution

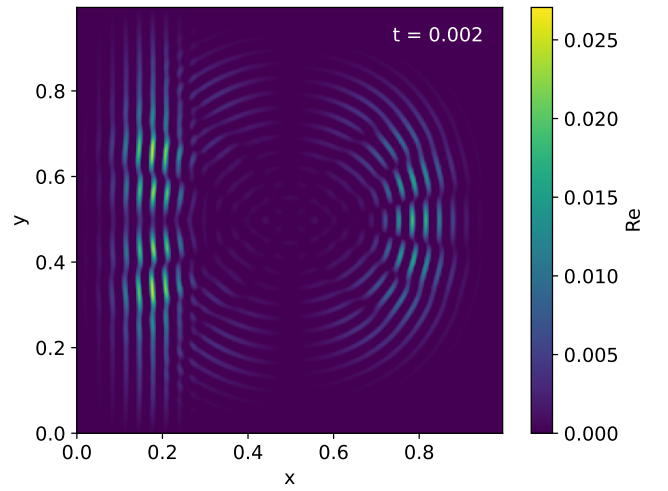
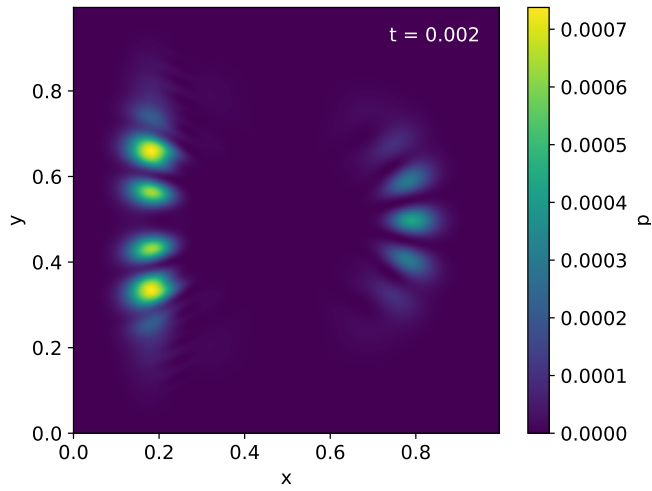
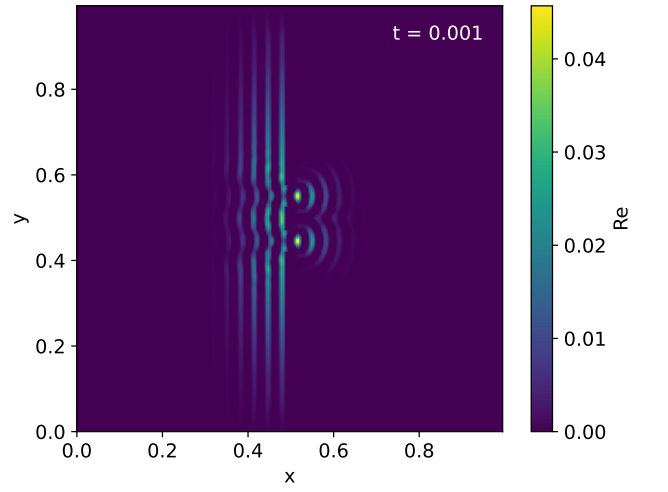
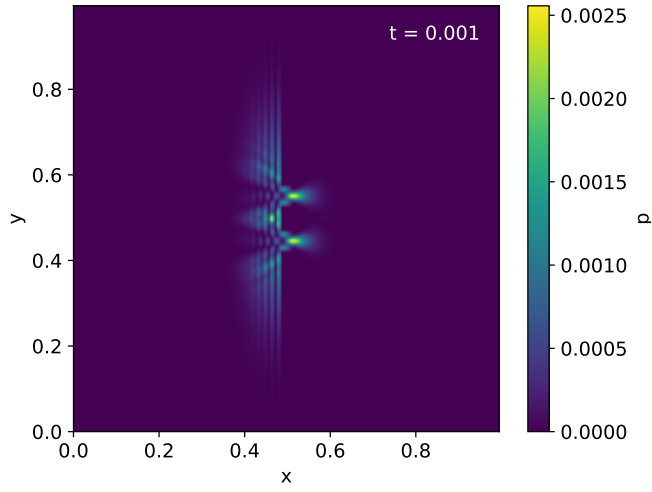
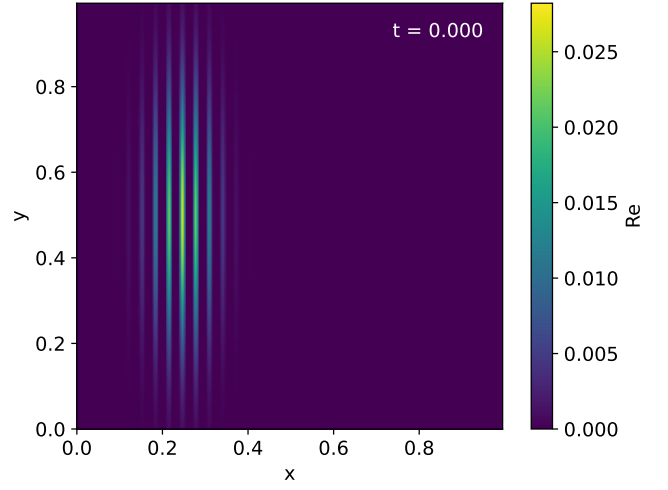
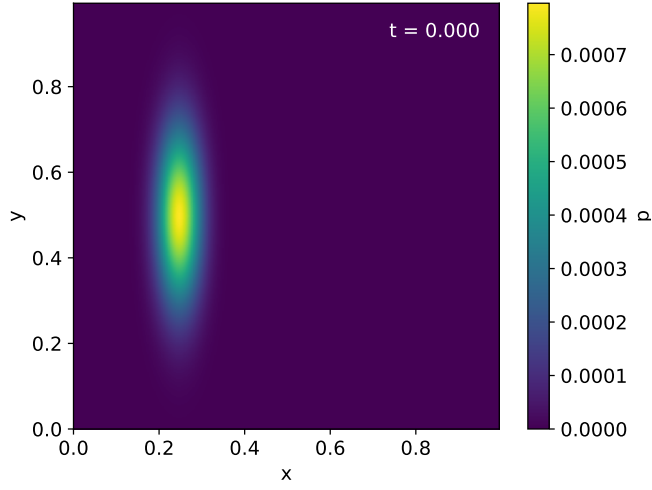


FIG. 2. Colormap plots that illustrate the time evolution of the 2D probability function, at times $t = 0$, $t = 0.001$ and $t = 0.002$ respectively.

FIG. 3. Colormap plots that show $\text{Re}(u_{ij})$, at times $t = 0$, $t = 0.001$ and $t = 0.002$ respectively.

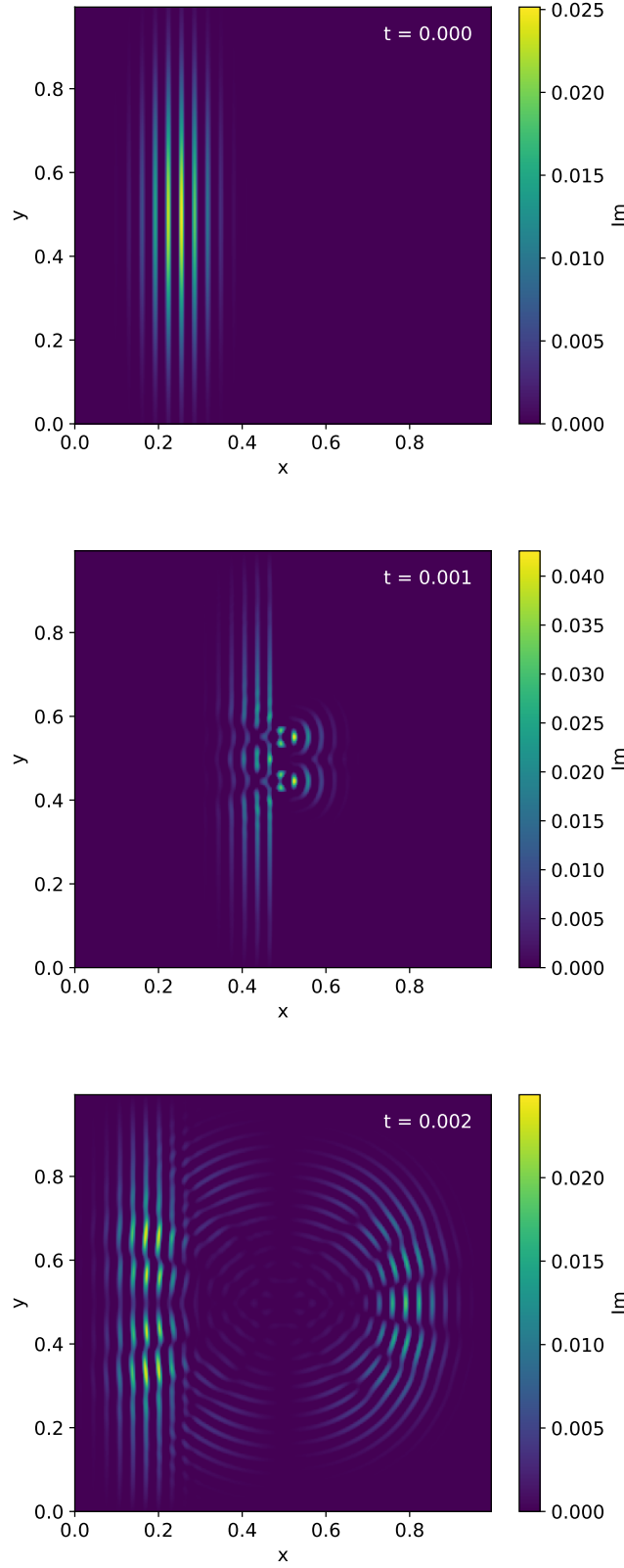


FIG. 4. Colourmap plots that show $\text{Im}(u_{ij})$, at times $t = 0$, $t = 0.001$ and $t = 0.002$ respectively.

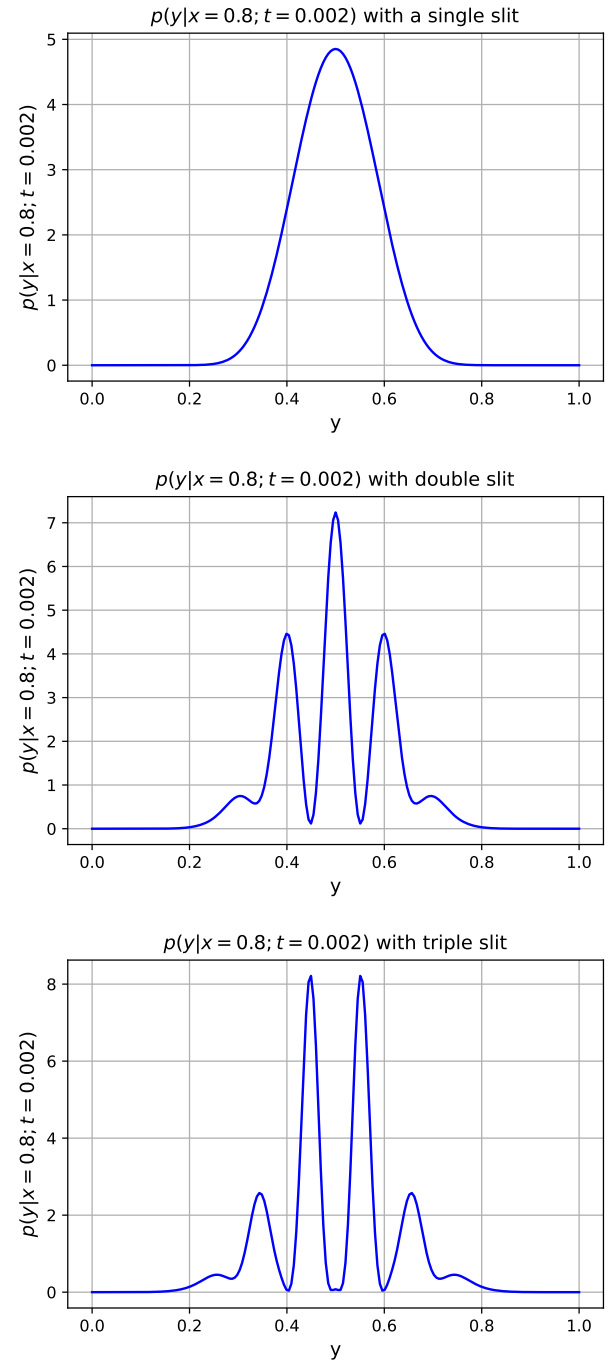


FIG. 5. Plots that illustrate the 1D probability function spanning the entire y axis at a position $x = 0.8$ and at a certain time $t = 0.002$ for the cases with 1, 2 and 3 slits respectively.

which illustrates the desired phenomenon (a particle which follows a wave function, passing through a n -slits configuration).

After that, we have studied a double slit configuration (Figure 2). It has been observed how the probability function of detecting the particle at a specific position

(x, y) changes as time goes through. For this simulation we have also looked at how the real and imaginary parts of the wave function change with time since in these plots (Figures 4 and 3) it can be seen in a more clear way the behaviour of the wavefronts before, while and after going through the slit. Observing them, one can get a significantly good look at how the expected diffraction and reflection processes take place.

Finally, for the already mentioned configuration and also for one and three slits, it has been studied the detection probability when spanning the entire y axis at a certain x position and time. The diffracted intensity is essentially zero except in the directions where the waves coming out from all the slits adds up in phase, and this explains what it is observed in Figure 5 since the diffracted intensity is proportional to the probability density.

A possible improvement in the experiment could be to simulate a thinner wall without slits and try to illustrate the tunnel effect, in which a quantum particle “crosses” the potential barrier so, if the wall is thin enough, some significant parts of the probability function might reach the other side of the wall.

V. APPENDIX

Discretization of Schrödinger's equation

Starting from Equation 1, using Taylor's expansion for the second derivative, where

$$\frac{\partial^2 f(x)}{\partial x^2} \Big|_{x=x_0} = f''(x_0) = \frac{f_{x_0+\Delta x} - 2f_{x_0} + f_{x_0-\Delta x}}{(\Delta x)^2} + O(\Delta x^2),$$

and using the Crank-Nicolson approach, we will get:

$$\begin{aligned} i \frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t} = & - \left(\frac{u_{i+1,j}^{n+1} - 2u_{i,j}^{n+1} + u_{i-1,j}^{n+1}}{2(\Delta x)^2} - \frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{2(\Delta x)^2} \right) \\ & - \left(\frac{u_{i,j+1}^{n+1} - 2u_{i,j}^{n+1} + u_{i,j-1}^{n+1}}{2(\Delta y)^2} - \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{2(\Delta y)^2} \right) \\ & + V_{i,j} \frac{u_{i,j}^{n+1} - u_{i,j}^n}{2} \end{aligned}$$

Like $\Delta x = \Delta y$, we will rename those variables, so, now, $\Delta x = \Delta y = h$. Then, we get:

$$\begin{aligned} i \frac{u_{i,j}^{n+1} - u_{i,j}^n}{\Delta t} = & - \left(\frac{u_{i+1,j}^{n+1} - 2u_{i,j}^{n+1} + u_{i-1,j}^{n+1}}{2h^2} - \frac{u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{2h^2} \right) \\ & - \left(\frac{u_{i,j+1}^{n+1} - 2u_{i,j}^{n+1} + u_{i,j-1}^{n+1}}{2h^2} - \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{2h^2} \right) \\ & + V_{i,j} \frac{u_{i,j}^{n+1} - u_{i,j}^n}{2} \end{aligned}$$

Then, defining r as $r = \frac{i\Delta t}{2h^2}$, we can rewrite the Crank-Nicolson approximation of Equation 1 as

$$\begin{aligned} u_{i,j}^{n+1} - r(u_{i+1,j}^{n+1} - 2u_{i,j}^{n+1} + u_{i-1,j}^{n+1}) - r(u_{i,j+1}^{n+1} - 2u_{i,j}^{n+1} + u_{i,j-1}^{n+1}) \\ + \frac{i\Delta t}{2} V_{i,j} u_{i,j}^{n+1} = u_{i,j}^n - r(u_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n) \\ - r(u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n) - \frac{i\Delta t}{2} V_{i,j} u_{i,j}^n \end{aligned}$$

where $r \equiv \frac{i\Delta t}{2h^2}$.

This equation can be expressed as a matrix equation such as $Au^{n+1} = Bu^n$, where A and B are matrices with coefficients a_i (for matrix A) and b_i (for matrix B) in the main diagonal, with r (for matrix A) and $-r$ (for matrix B) in some spots, filling the rest of each matrix with 0. This is

$$A = \begin{bmatrix} a_0 & -r & 0 & -r & 0 & 0 & 0 & 0 & 0 \\ -r & a_1 & -r & 0 & -r & 0 & 0 & 0 & 0 \\ 0 & -r & a_2 & 0 & 0 & 0 & -r & 0 & 0 \\ -r & 0 & 0 & a_3 & -r & 0 & -r & 0 & 0 \\ 0 & -r & 0 & -r & a_4 & -r & 0 & -r & 0 \\ 0 & 0 & -r & 0 & -r & a_5 & 0 & 0 & -r \\ 0 & 0 & 0 & -r & 0 & 0 & a_6 & -r & 0 \\ 0 & 0 & 0 & 0 & -r & 0 & -r & a_7 & -r \\ 0 & 0 & 0 & 0 & 0 & -r & 0 & -r & a_8 \end{bmatrix}$$

and

$$B = \begin{bmatrix} b_0 & r & 0 & r & 0 & 0 & 0 & 0 & 0 \\ r & b_1 & r & 0 & r & 0 & 0 & 0 & 0 \\ 0 & r & b_2 & 0 & 0 & 0 & r & 0 & 0 \\ r & 0 & 0 & b_3 & r & 0 & r & 0 & 0 \\ 0 & r & 0 & r & b_4 & r & 0 & r & 0 \\ 0 & 0 & r & 0 & r & b_5 & 0 & 0 & r \\ 0 & 0 & 0 & r & 0 & 0 & b_6 & r & 0 \\ 0 & 0 & 0 & 0 & r & 0 & r & b_7 & r \\ 0 & 0 & 0 & 0 & 0 & r & 0 & r & b_8 \end{bmatrix}$$

for 9x9 matrices, where $a_k = 1 + 4r + \frac{i\Delta t}{2} V_{ij}$ and $b_k = 1 - 4r - \frac{i\Delta t}{2} V_{ij}$

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